

RAÚL DE LA FUENTE

[EMAIL](#) | [WEBSITE](#) | [GITHUB](#) | [LINKEDIN](#)

URBANA, IL

SUMMARY

Second-year Computer Science Ph.D. student at UIUC, advised by Prof. Josep Torrellas. I work on microarchitecture for 3D-stacked CPUs, GPUs, and accelerators. Seeking a Summer 2027 computer architecture internship.

EDUCATION

University of Illinois Urbana-Champaign | GPA: 3.75/4.0 Fall 2025 - Present
Ph.D. | Computer Science Urbana, IL
Advisor: Prof. Josep Torrellas

University of Chile | GPA: 7.0/7.0 Fall 2023 - Spring 2024
M.S. | Computer Science Santiago, Chile

University of Chile | GPA: 6.4/7.0 Fall 2018 - Fall 2022
B.Eng. | Computer Science and Engineering Santiago, Chile

RELEVANT COURSEWORK

Fall 2026: CS598APE - Advanced Performance Engineering, CS598HSA - Hardware & Software for AI
Spring 2026: CS533 - Parallel Computer Architecture, ECE425 - VLSI Design
Fall 2025: CS433 - Computer System Organization, CS426 - Compiler Construction, CS484 - Parallel Programming

RESEARCH EXPERIENCE

I-acoma Group, UIUC Urbana, IL
Graduate Research Assistant | Advisor: Prof. Josep Torrellas August 2025 - Present

- Improving performance of distributed collectives on hopper-like GPGPU clusters through 3D NoCs.
- Contributing to a universal LLM accelerator that maps kernel sub-functions onto logic and memory chiplets.

Dept. of Electrical Engineering, University of Concepción Concepción, Chile
Research Assistant | Advisor: Prof. Luciano Radrigan July 2022 - December 2024

- Wrote C/C++ firmware for TinyML IoT nodes with network-aware switching between edge and cloud inference.

CLCERT Group, University of Chile Santiago, Chile
Research Assistant | Advisor: Prof. Alejandro Hevia January 2022 - December 2024

- Migrated a homomorphic encryption system from finite-field to elliptic curve cryptography, cutting computation cost and enabling distributed key generation to run on resource-constrained embedded devices.

AIO Team, INRIA Paris, France
Research Intern | Advisor: Prof. Thomas Watteyne January 2023 - April 2023

- Evaluated the performance and energy consumption of a satellite IoT module; published as a poster at IPSN '23.

TECHNICAL SKILLS

Programming: C/C++ | CUDA | Python | Rust | RISC-V | GoLang | Java | Scala | Bash
Software: Docker | Git | TensorFlow | Pytorch | LLVM IR | MPI | OpenMP | Valgrind
Hardware: Vortex | SystemVerilog | Gem5 | Ramulator | Virtuoso | Innovus | PrimePower | DesignCompiler

PUBLICATIONS

Enhancing Predictive Maintenance in Mining Mobile Machinery through a TinyML-enabled Hierarchical Inference Network. Raúl de la Fuente, Luciano Radrigan and Anibal S Morales. In *IEEE Access* (IEEE Access).

Poster abstract: Hands-on Evaluation of Kinéis Satellite IoT Technology. Raúl de la Fuente and Thomas Watteyne. In *Proceedings of the 22nd International Conference on Information Processing in Sensor Networks (IPSN '23)*.